Introduction

What Machine Learning is

ML (= Machine Learning) is a **field of Artificial Intelligence**, concerning **algorithms** that can:

- **Learn from data** → the system **learns patterns** from data
- **Generalize to unseen data**
- **Perform tasks without explicit instructions** → the **behavior is learned**, not manually coded

The idea behind ML

The idea behind ML follows a set structure:

- **Knowledge is acquired** through algorithms **by learning** from data
- **Knowledge is represented by a model**
- The **model** is **used on future data**

ML applications

ML is used to:

- Recognize and generate patterns
- Recognize anomalies
- Predict events and best moves

Training and Testing phases

The ML process is structured in 2 main parts:

- **Training**: **creation of the model from** the observation of **existing data**
- **Testing**: **prediction** based on the previously generated model

ML as a triplet: experience, class of tasks, performance

The **more the experience** E **in a class of tasks** T, the **more the performance** P of the algorithm

- A well-defined learning task is given by a **triplet** $< T, P, E >$

Artificial Intelligence vs Machine Learning vs Deep Learning

The 3 above are not the same:

- **AI** : enables machines to **perform tasks** which typically require **human intelligence**
- **ML**: subset of AI, **systems automatically learn patterns** and **make predictions**
- **DL**: subset of ML, **relays on** multi-layered **neural networks**